

Make-up for Seatwork #1

Quantitative Methods

Midterm

- I. Determine if the given problems are MPC, permutation and combination, then solve it. Show all solutions and box the final answer.
1. A group of 15 athletes are competing for 100-m dash. In how many ways can we have gold, silver, and bronze medalists?
 2. Listing your 3 favorite desserts, in order, from a menu of 10. How many possible sets of desserts you can form?
 3. How many four-digit numbers can be formed using the non-zero digits if repetition is not allowed?
 4. How many combinations of 5 records can be chosen from 12 records offered by a record club?
 5. A serial number consists of seven characters. The first three characters are uppercase letters of the alphabet. The next two are selected from the digits 1 to 9. The last two are uppercase letters of the alphabet. How many serial numbers are possible if no letter or number can be used twice in the same serial number?
 6. Suppose there are 12 distinct points on a circle. How many different triangles can be formed with the vertices at the given points?
- II. Solve the following and describe whether it is impossible, unlikely, even chance, likely or certain. Show the complete solution and box your final answer.
1. What is the probability of getting a head when tossing a coin?
 2. A school has 820 male students and 835 female students. If a student is selected at random, what is the probability that the student would be a female?
 3. A bowl of candies contains 3 blue, 1 yellow, 1 red, 1 green, and 2 black. If a child picks one at random, what is the probability that he will pick blue?
 4. What is the probability of getting a king of hearts from a complete pack of cards?
 5. What is the probability of obtaining an even number when a die is rolled?
 6. What is the probability of the sun rising in the east?
 7. What is the probability of encountering a real unicorn?